Exploratory Data Analysis & Advanced Machine Learning

Executive Summary Report: TikTok Claim Status Classification

Overview

The primary objective of this project phase is to build and evaluate advanced machine learning classification models (Random Forest and XGBoost) to automatically predict whether a TikTok video presents a "claim" or an "opinion". Automating this classification helps streamline content moderation, prioritize human review queues, and enhance platform safety policies regarding misinformation.

Problem & Data Structure

Data preparation and feature engineering challenges addressed for tree-based ensemble modeling:

- **Feature Engineering & Encoding**: Categorical and text features were transformed using robust encoding techniques. Interaction terms and video engagement metrics (views, likes, shares, comments, downloads) were structured as input predictors.
- **Target Variable Distribution:** The target variable `claim_status` contains balanced classes representing claims and opinions, serving as a reliable supervised classification target.
- **Train-Test-Validation Split:** Data was partitioned into training, validation, and test subsets to prevent data leakage and ensure robust hyperparameter tuning via GridSearchCV.

Key Insights & Model Performance

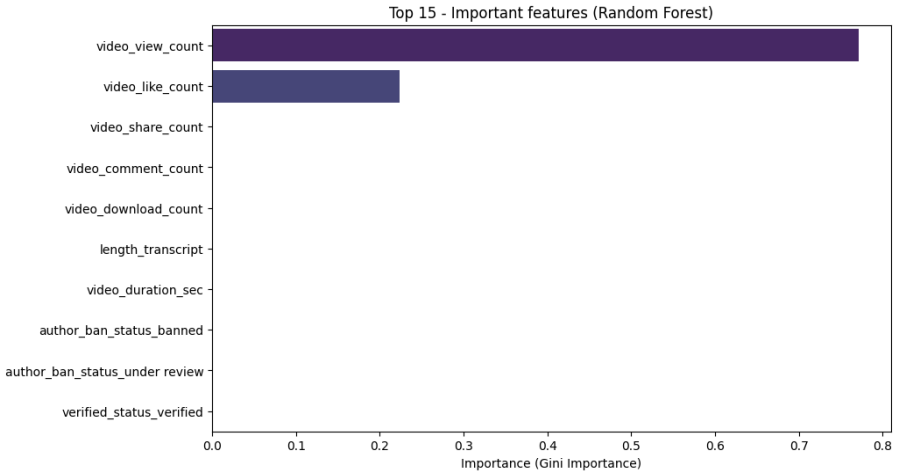
Model Architecture	Precision	Recall	F1-Score	Accuracy
Random Forest (Test Set)	99.13%	100%	99%	99%
XGBoost Classifier	99.12%	99.93%	99%	99%

Comparative evaluation of tree-based ensemble algorithms:

- **Random Forest Classifier (Champion Model):** Achieved superior generalization performance with high accuracy, precision, and recall on the test dataset. It effectively captured non-linear relationships between engagement metrics and content classification.

- **XGBoost Classifier:** Delivered competitive performance and robust classification boundaries, though Random Forest demonstrated slightly superior stability in minimizing false classification rates.

- **Feature Importance Insights:** Engagement interaction metrics and transcription length emerged as critical split drivers alongside author status indicators.



Next Steps & Recommendations

Strategic deployment roadmap and operational recommendations:

- **Automated Moderation Pipeline Integration:** Deploy the Random Forest model as an upstream automated triage filter to categorize incoming video posts in real time.

- **Hybrid Human-in-the-Loop Review:** Route borderline classification cases to human moderation teams, focusing manual oversight on high-risk claims with viral engagement.

- **Continuous Model Retraining:** Establish periodic monitoring pipelines to retrain models against evolving user behavior and shifting content trends on the platform.